

Saiakhil Chilaka

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EDUCATION

Cornell University, College of Arts & Sciences

Ithaca, NY

B.A. in Computer Science

Expected May 2027

- **Relevant Courses:** Data Structures & Algorithms, Functional Programming, Linear Algebra, Calculus 3, Discrete Math

EXPERIENCE

Technology Implementation Associate

September 2025 – Present

Cornell Data & Strategy

Ithaca, NY

- Built a multi-class ML classifier for Albany Med Health System to resolve ambiguous payer records which make up 37.6% of AMC hospital system's discharges, enabling the first reliable estimate of true government payer exposure in the market.
- Trained a LightGBM gradient boosting regressor on a dataset of 100k records to forecast market gaps across competitors.
- Predicted demand for Sciencecenter's \$12.3M of assets through Markov transition probabilities on a location graph.

SWE + AI Intern

December 2025 – Present

PersonaAI

Remote

- Built a multi-stage GPT-4 pipeline that synthesizes deep psychological personas from a single prompt, then simulates real-time interviews with context-aware trait injection to replicate human respondent behavior.
- Engineered a serverless backend across 20+ Supabase Edge Functions with a 3-stage LLM pipeline (trait synthesis, summary compression, & DALL-E image generation) with dynamic context selection to reduce per-request token overhead.

Data Analyst Intern

April 2024 – Present

Vanasse Hangen Brustlin, Inc. (VHB)

Remote/Raleigh, NC

- Wrote 6 scripts in R integrating roadway & Census data to identify gaps in NC Dept. of Transportation data.
- Built a Python automation that parses crash reports, geocodes crash locations, autoclassifies through Claude LLM.

Research Intern

February 2024 – December 2024

NC State University

Raleigh, NC

- Developed a CNN ensemble framework for Multiple Sclerosis diagnosis on embedded systems under Dr. Zishan Guo using knowledge distillation; achieved 97.5% accuracy & 97.8% AUC-ROC; Published paper as First Author in IEEE Xplore.
- Built a fairness-aware skin cancer classification framework for embedded devices with knowledge distillation, Generative Adversarial Networks, & Gaussian white noise, improving fairness by 37%, accuracy by 19%, & inference latency by 32%.

Research Intern

May 2024 – November 2024

Duke University

Durham, NC

- Used NLP & Latent Dirichlet Allocation to model topic distributions across citation networks of up to 5 research papers. Tuned LDA hyperparameters to achieve 0.71 topic coherence & 0.88 diversity across tech & medical fields

PROJECTS

Cognitive Load Mapping | *Python, MediaPipe, OpenCV, ResNet50*

January 2026 – Present

- Built a model that infers real time mental fatigue through tracking eye movement, blink patterns, and head pose at 30fps
- Engineered a BiLSTM + 4-head Attention pipeline (gaze, fixation, EAR, head pose) over 60-step windows of a 24-dim feature vector with a FiLM adapter for zero-shot per-user personalization, achieving 30ms CPU inference on Apple Silicon
- Designed a self-supervised calibration system fitting a RANSAC homography from natural reading saccades, no calibration required, mapping L2CS-Net gaze vectors to screen coordinates in 60–120s to enable live per-paragraph load heatmapping

FlavorNet | *Python, PyTorch, SQLite, Streamlit, Claude API*

December 2025 – Present

- Architected a Graph Neural Network that discovers non-obvious food ingredient pairings using molecular flavor data, recipe co-occurrence, and multimodal features with InfoNCE contrastive loss over 400k+ recipes & 1k+ ingredients.
- Trained a 3-layer heterogeneous GAT on 436k ingredient pairs using dual supervision to score pairings by compatibility & rarity with an active learning loop sampling the 100 most uncertain pairs, collecting ratings, and fine-tuning in 10 epochs.
- Designed a data pipeline scraping 3 databases with incremental SQLite caching, reducing re-fetch overhead by 90%.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, TypeScript, R, Julia, HTML + CSS, SQL

Frameworks & Libraries: React, Electron, FastAPI, Recharts, PyTorch, TensorFlow, scikit-learn, XGBoost, pandas

Databases & Infrastructure: Git, PostgreSQL, Redis, Supabase, Celery